

Kaan Baha Sever

SOFTWARE DEVELOPER | MATH-DRIVEN SOLUTIONS & ALGORITHMS

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Software developer with a mathematics background (Istanbul University). I build systems software in modern C++ and Go: desktop agents, telemetry and device-management software, and algorithm-driven applications, together with the CI/CD pipelines that test, package and ship them. From 2021 to 2024 I owned the development lifecycle at crowd.inc, from database schema to Linux servers.

EXPERIENCE

Software Developer · crowd.inc Jul 2021 – Mar 2024

crowd.inc was a platform where people shared their ideas and found help for them. It had hundreds of users and was home to many projects, ideas and goals; later we turned towards private ideas. I owned the software development lifecycle end to end, from relational schema design in PostgreSQL to RESTful APIs in Python and Flask with a jQuery front end.

- Designed granular role-based access control (RBAC) with strict permission boundaries between public and private data.
- Built scalable feed pagination and dynamic data loading across the API and the front end.
- Provisioned and configured the Linux servers the platform ran on.
- Kept the system reliable through rigorous unit and integration testing.

Python · Flask · PostgreSQL · jQuery · Linux

EDUCATION

Bachelor of Science in Mathematics · Istanbul University · Faculty of Science Nov 2019 – Present

- **Rocket and Space Club, Vice President (2019–2022):** developed avionics, telemetry, ground-control and flight-simulation software for the team's three rockets, one low-altitude (5,000 ft) and two high-altitude (10,000 ft); wrote and tested Kalman filters for the sensor data and used PID controllers in vertical-landing experiments.
- **Mathematics Club (later university years):** organised and ran academic events, such as seminars and logic and mathematics competitions, for a community built around theoretical discussion.
- **Google Developer Student Clubs (GDSC), Core Team Member (2023):** hosted technical workshops and live streams on Flask, HTML and Git/GitHub for the community, and organised Cyber Security Week.

SELECTED PROJECTS

Asion · asion.app Since 2024 · in development, early access

A privacy-first, cross-platform productivity and workstation activity tracker for individuals and teams, and for academic time tracking, built with self-hosting in mind. It runs on native OS event hooks on macOS, Linux and Windows, with a lightweight daemon architecture, gRPC/Protobuf IPC and an encrypted SQLCipher store. I built its multi-platform GitHub Actions pipelines, with cross-compilation runners, for one-click testing, packaging, artefact generation and deployment.

C/C++ · Objective-C · Go · gRPC

Novacast · novacast.app Since 2025 · in production

Software that manages many screens of different makes from one web panel, on Windows, macOS, Linux, Samsung TV, LG TV and Android TV. We built it at a computer firm where I worked to replace bloated, inadequate software: a Rust player on each screen, reached by a Go server through an MQTT broker.

Rust · Go · MQTT

Karecik · karecik.com Since 2021 · in production · GPL-3.0

A multi-tenant QR-menu SaaS used by local cafés and restaurants. It began as a QR menu for a friend's café, and he now uses it in all three of his branches. Built for zero downtime and low latency. Early prototypes date from 2021 under earlier working titles; it shipped to production in 2026.

Go · PostgreSQL

Açık Matematik · acik-matematik.com Since 2023 · in production · open source

An academic publishing platform and modern textbook initiative for undergraduate mathematics in Turkish, with reproducible numerical computing. Conceived in 2023 and launched in June 2026; the open-source release was announced in September 2026.

Quarto · Markdown · Python

SKILLS

Programming languages	C++ (C++23) · Go · C · Python · C# · TypeScript / JavaScript
Systems & network programming	gRPC · Protobuf · MQTT · OS event hooks · Linux systems
DevOps & tools	Git · GitHub · GitHub Actions · CI/CD pipelines · Cross-compilation · Bash · Batch
Databases	PostgreSQL · SQLite (SQLCipher) · MySQL · MS SQL
Web technologies	React · Astro · Flask · REST APIs · HTML/CSS
Scientific computing	MATLAB · Mathematica · R

RESEARCH

Autonomous parachute guidance · github.com/KaanBahaSever/AutonomousParachute

A trajectory-steering algorithm in Python that uses linear algebra and atmospheric descent dynamics to guide a payload to a designated landing coordinate.

High school research project · TÜBİTAK (Scientific and Technological Research Council of Türkiye) 2019
– Researched cryptography and visual programming logic, including an early prototype of a flowchart-based visual coding tool.

CERTIFICATES & ACTIVITIES

21st Century Competencies · YetGen · Mehmet Zorlu Foundation 2020

Erasmus+ Youth Exchange · Arrecife, Spain Jun 2017 – Dec 2018

LANGUAGES

Turkish (native) · English